Zhonglin Liu

PhD in Mathematics | The University of Hong Kong | HKPFS Awardee
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Profile

Mathematics PhD researcher working at the intersection of mathematical finance, quantitative risk management, and digital-asset infrastructure.
Research focus: dynamic risk-sharing pool expansion, admission and contribution pricing, tokenised settlement, stablecoin reserve/run risk, and DeFi risk governance. Combines theoretical modelling with Python implementation. Seeking a Summer 2027 off-cycle research project in tokenised finance, digital-asset risk, or quantitative risk.

Education

The University of Hong Kong **2026–2030 (expected)**
PhD in Mathematics | Hong Kong PhD Fellowship Scheme (HKPFS) Awardee
Official supervisor: Prof. Wai-Ki Ching; working supervisor: Prof. Alfred Chong
Research: dynamic risk sharing and pooling; admission, risk pricing and participation constraints.

The University of Hong Kong **2022–2026**
BSc in Mathematics and Economics, First Class Honours | GPA 3.86/4.3
Dean's Honours List (multiple years); C.V. Starr Scholarship; Wong Yung Chow Prize in Mathematics.
Relevant coursework: Probability, Numerical Analysis, Optimization, PDEs, Game Theory, Corporate Finance.

National University of Singapore **Jan–May 2025**
HKU Worldwide Undergraduate Exchange Programme | Semester GPA 4.9/5.0

Research and Quantitative Projects

Dynamic Risk-Sharing and Pool Expansion **Current**
Modelling how risk-sharing pools expand over time: participant admission, entry pricing, common reserve ownership, buy-in, loss allocation, and participation constraints. Methods include risk measures, BSDEs, stochastic control, dynamic optimisation, and game theory.

Mechanism Design for Improving LLMs **2025**
Designed incentive-compatible reward–penalty and verification mechanisms; used optimisation, game theory, and simulation to study platform and participant outcomes. Manuscript under review at TMLR.

Machine Learning in Online Portfolio Selection **2023–2024**
Built LSTM and XGBoost pipelines for asset-price prediction and portfolio-selection strategies; published in peer-reviewed conference proceedings.

Selected Publications

- Liu, Z., Keppo, J., Mandal, M., & Tan, H. M. (2026). "Screening Feedback for Language Models with Costly Verification." *Transactions on Machine Learning Research* — under review.
- Liu, Z., Zhao, Y., Lyu, B., & Ching, W.-K. (2025). "Integrating Machine Learning Methods for the Prediction in Online Portfolio Selection Problems." *International Conference on Applied Research in Business, Management and Economics.*

Technical Skills

Programming: Python (scientific computing, simulation), MATLAB, LaTeX.
Quantitative: risk measures, BSDEs, stochastic control, convex duality, game theory, Monte Carlo, numerical optimisation.
Finance / DeFi: derivatives pricing, portfolio selection, risk sharing and pool expansion, stablecoin/reserve risk, risk governance.

Awards and Activities

HKPFS; First Class Honours; C.V. Starr Scholarship; Wong Yung Chow Prize in Mathematics; Dean's Honours List.
Student Teaching Assistant, HKU Mathematics (2023–2024); created educational mathematics videos using Manim, strengthening communication and technical presentation skills.

Availability

Summer 2027; approximately 8–12 weeks. Singapore or Hong Kong, subject to team needs and applicable work-pass requirements.